Energy-Efficient Electric Vehicle Routing with Stochastic Congestion and Discrete Partial Charging

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Abstract—This paper presents the design, implementation, and evaluation of an end-to-end system for energy-efficient Electric Vehicle (EV) routing under stochastic, time-dependent traffic conditions. The problem requires jointly optimising the delivery sequence, speed profile, and charging decisions for an EV serving five delivery locations across a real urban road network, subject to battery feasibility and a deadline constraint. The work is formalised as a finite-horizon Markov Decision Process (MDP) and solved under two separate optimisation objectives: Subtask A minimises the number of charging stops, and Subtask B minimises the total monetary cost of charging. The system integrates three real-world datasets—an OpenStreetMap road network of South Los Angeles, six days of Caltrans PeMS freeway speed data, and a Kaggle EV charging station dataset—into a unified simulation environment. A Dueling Deep Q-Network (DQN) agent with warm-start guided exploration is trained for each subtask and benchmarked against a Dijkstra nearest-neighbour baseline. The Dijkstra baseline achieves 100% task completion with a single mandatory charging stop, consuming approximately 14.1 kWh per episode. After 3 000 training episodes, the DQN agents achieve completion rates of 82% (Mode A) and 75% (Mode B). The best greedy evaluation episode for Mode A completes the full tour in 7 hops with only 1 charging stop; Mode B completes in 9 hops with 3 stops spread across cheaper AC Slow stations, demonstrating cost-aware behaviour. These results are grounded in a theoretically rigorous MDP formulation and a nonlinear charging model adopted from recent published work by members of the same laboratory.

Index Terms—Electric vehicle routing, Markov decision process, deep reinforcement learning, dueling DQN, stochastic congestion, discrete partial charging, PeMS traffic data, OpenStreetMap.

1 Introduction and Motivation

The global transition toward electric mobility is accelerating rapidly. Major logistics providers—including DHL, FedEx, JD Logistics, and Amazon are integrating EVs into their last-mile delivery fleets to meet regulatory emission targets and reduce long-run operational costs. However, EV adoption in logistics introduces challenges absent from Internal Combustion Engine (ICE) fleets: limited driving range, time-consuming nonlinear recharging, sparse charging infrastructure, and strong coupling between routing decisions and energy feasibility.

The classical Vehicle Routing Problem (VRP) asks for the optimal delivery sequence minimising total distance or time. The Electric Vehicle Routing Problem (EVRP) extends VRP by additionally requiring the vehicle to maintain positive battery charge and to optionally detour to charging stations. Unlike refuelling an ICE vehicle, EV charging is nonlinear—batteries charge rapidly from 0 to 80% State of Charge (SoC) but slow down significantly beyond that threshold, making charging duration a complex function of arrival charge level and charger type.

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- Supervised by Prof. Arnab Sarkar, Advanced Technology Development Centre (ATDC), IIT Kharagpur, and Dipankar Mandal, Department of Artificial Intelligence, IIT Kharagpur.

- Course: AI69002 Design Laboratory I. Submitted April 30, 2026.

This project addresses an EVRP variant that is more general than most existing work in two key ways. First, the route is not pre-determined; both the delivery sequence and the charging decisions are made jointly as part of the optimisation. Second, road speeds are modelled as stochastic, time-dependent random variables derived from real traffic sensor data. Furthermore, the agent controls its speed on each segment, creating a three-way coupling between routing, speed, and charging decisions.

Two separate optimisation objectives are studied:

- **Subtask A:** Minimise the total number of charging stops. Relevant where charging infrastructure is scarce or driver dwell time is the binding constraint.
- **Subtask B:** Minimise the total monetary cost of charging. Relevant where cheaper, slower chargers should be preferred over expensive fast chargers when time permits.

Both subtasks share the same environment, state space, and transition dynamics; only the reward signal differs.

2 Related Work

2.1 Vehicle Routing Problem

The VRP is one of the most extensively studied problems in combinatorial optimisation [11]. The Capacitated VRP (CVRP) enforces vehicle load constraints. Both are NP-Hard, requiring heuristics or metaheuristics for practical problem sizes.

2.2 Electric Vehicle Routing Problem

The EVRP introduces battery capacity, energy consumption, and charging constraints on top of classical VRP. Two primary solution streams exist. The *monolithic* approach solves routing and charging jointly. The *hierarchical two-step* approach, validated by Montoya *et al.* [2], first generates routes with CVRP algorithms and then solves the Fixed Route Vehicle Charging Problem (FRVCP) to make each route energy-feasible.

2.3 Mandal *et al.* (2025) — Direct Precursor

The most directly relevant prior work is Mandal, Sarkar, and Mondal [1], *"A Discrete Partial Charging Enabled Dynamic Programming Strategy for Optimal Fixed-Route Electric Vehicle Charging,"* published in ACM Transactions on Embedded Computing Systems (Vol. 24, No. 5s, Article 154). Both first authors are affiliated with IIT Kharagpur—the same institution as this project's supervisor—making this the direct intellectual precursor of the present work.

Their paper introduces **FRVCP-DPC** and proposes the **BEFRG** algorithm: a recursive DP with memoisation that jointly selects detour points, charging stations, and discrete charge levels to minimise total route time. Its time complexity is $\mathcal{O}(|R| \cdot |V_C| \cdot \Gamma \cdot \gamma)$. The piecewise-linear nonlinear charging model, load-dependent energy formula, and discrete battery discretisation are adopted directly in this project.

This project extends Mandal *et al.* in three fundamental ways: (i) the route is a decision variable, not a fixed input; (ii) speed is controllable and stochastically constrained per segment; (iii) road speeds follow time-indexed probability distributions from real traffic data, rather than deterministic or instantaneous values.

2.4 Reinforcement Learning for Routing

RL has been applied to EVRP with increasing success [3]. The Dueling DQN architecture [4] separates state value estimation from action advantage estimation, improving stability in problems where state quality dominates individual action value— exactly the case in routing, where completing all deliveries matters far more than the specific next node chosen.

3 Problem Statement

Given a directed urban road network with a fixed depot, five delivery locations, and five heterogeneous EV charging stations, an Electric Vehicle with battery capacity $B = 12\,\text{kWh}$ must visit all delivery locations and return to the depot within deadline $T_{\max} = 18{:}00$, starting at $t_0 = 08{:}00$ with a full battery. Road speeds on each segment follow time-dependent stochastic distributions derived from real PeMS sensor data. The vehicle consumes energy as a function of its chosen speed, current cargo load, and prevailing congestion. Energy is also lost at each intersection. The vehicle may optionally recharge to one of a finite set of discrete charge levels following a nonlinear piecewise-linear charging curve.

The agent must jointly decide: (1) which key node to visit next; (2) what speed bin to travel at; (3) whether to charge upon arrival at a charging station; (4) if charging, up to which discrete charge level.

Subtask A: Minimise total charging stops, subject to battery feasibility and $t_{\text{return}} \leq T_{\max}$.

Subtask B: Minimise total charging cost ($), subject to the same constraints.

The worst-case tour energy at rush hour and medium speed (40 km/h) is approximately 17.5 kWh, exceeding the 12 kWh battery capacity—making at least one charging stop mandatory on every episode. The optimisation question is therefore not *whether* to charge, but **when, where, and how much**.

4 Mathematical Model

4.1 Graph Model

The urban road network is modelled as a directed graph $G = (V, E)$. An **abstract routing graph** is constructed over a reduced set of key nodes:

$$V^* = \{v_0\} \cup V_D \cup V_C \tag{1}$$

where v_0 is the depot, $V_D = \{D_1, \ldots, D_5\}$ are delivery nodes, and $V_C = \{C_1, \ldots, C_5\}$ are charging stations. All pairwise shortest paths between elements of V^* are precomputed using Dijkstra's algorithm, yielding: $\delta^*(u, v)$ (distance in km) and $\Pi(u, v)$ (full OSMnx node sequence, used for visualisation).

This abstraction reduces the effective action space from thousands of graph neighbours to at most $|V^*| - 1 = 10$ candidate next nodes per step.

4.2 Time-Dependent Stochastic Speed Model

Time is discretised into hourly slots $t \in \{0, \ldots, 23\}$. For each edge (i, j) and hour t, achievable speed is modelled as a truncated normal distribution:

$$v_{ij}(t) \sim \mathcal{TN}\big(\mu_{ij}(t),\, \sigma_{ij}(t),\, v_{ij}^{\min}(t),\, v_{ij}^{\max}(t)\big) \tag{2}$$

where $\mu_{ij}(t)$ and $\sigma_{ij}(t)$ are estimated from PeMS 5-minute observations, and $v^{\min}, v^{\max}$ are the 5th and 95th percentile speeds for that sensor–hour pair. From the data, 4 992 unique (sensor, hour) pairs were estimated.

The agent selects a speed bin $s \in \{0, 1, 2\}$ (slow 20, medium 40, fast 70 km/h). The *realised speed* is:

$$v_{\text{actual}} = \text{clip}\big(V_{\text{bin}}[s] + \varepsilon,\, v_{ij}^{\min}(t),\, v_{ij}^{\max}(t)\big)\,, \quad \varepsilon \sim \mathcal{N}(0,\, 0.1\sigma_{ij}(t)) \tag{3}$$

Travel time on a leg: $\Delta t(u, v) = \delta^*(u, v)/v_{\text{actual}}$.

4.3 Energy Consumption Model

The energy model extends Mandal *et al.* [1] with speed-dependent drag and time-varying congestion.

Residual load before visiting node v_{k+1}:

$$L(v_{k+1}) = \sum_{v \in \mathcal{D} \cup \{v_{k+1}\}} \lambda(v), \quad \lambda(v) = 50\,\text{kg} \tag{4}$$

Load-dependent depletion rate [1]:

$$d_l(v_k, v_{k+1}) = \theta \times \big(L(v_{k+1}) + m_{\text{veh}}\big) \tag{5}$$

TABLE 1: Charging Station Parameters

Parameter	DC Fast	AC Slow
Breakpoints Z^c (kWh)	{0, 9.6, 11.2, 12.0}	{0, 9.6, 11.2, 12.0}
Breakpoints X^c (h)	{0, 0.55, 0.80, 1.10}	{0, 6.67, 7.78, 8.33}
Cost per kWh	\$0.35	\$0.15
Max rate	30 kW	7.2 kW

with $\theta = 0.0002\,\text{kWh}/(\text{kg km})$ and $m_{\text{veh}} = 1500\,\text{kg}$.

Road energy (with aerodynamic drag and congestion):

$$e_{\text{road}} = d_l \times \delta^*(u,v) \times \left(\alpha + \beta\, v_{\text{actual}}^2\right) \times \phi(t) \tag{6}$$

where $\alpha = 0.15\,\text{kWh/km}$ (rolling resistance), $\beta = 0.0004\,\text{kWh}\,\text{h}^2/\text{km}^3$ (aerodynamic drag), and:

$$\phi(t) = \begin{cases} 1.4 & t \in \{7,8,9,16,17,18\} \quad \text{(rush hour)} \\ 1.15 & t \in \{11,12,13\} \quad \text{(midday)} \\ 0.85 & t \geq 22 \text{ or } t \leq 5 \quad \text{(night)} \\ 1.0 & \text{otherwise} \end{cases} \tag{7}$$

Intersection crossing energy:

$$e_{\text{cross}} = n_{\text{cross}} \times 0.06\,\text{kWh}, \quad n_{\text{cross}} = \max\!\left(1,\ \lfloor \delta^*(u,v) \rfloor\right) \tag{8}$$

Total step energy and battery update:

$$e_{\text{step}} = e_{\text{road}} + e_{\text{cross}} \tag{9}$$

$$E_{t+1} = \max\!\left(E_t - e_{\text{step}},\ 0\right) \tag{10}$$

4.4 Nonlinear Piecewise-Linear Charging Model

The charging model is adopted directly from Mandal *et al.* [1]. Each station c is characterised by breakpoints $Z^c = \{z_0^c, \ldots, z_\omega^c\}$ (charge quantities, kWh) and $X^c = \{x_0^c, \ldots, x_\omega^c\}$ (cumulative time, h).

Charging rate in segment i:

$$\eta_i^c = \frac{z_i^c - z_{i-1}^c}{x_i^c - x_{i-1}^c} \tag{11}$$

For a charge from arrival SoC $y(c)$ to target $y'(c) = l \cdot ch$:

$$\alpha_i^c = \max\{y(c),\ z_{i-1}^c\}, \quad \beta_i^c = \min\{y'(c),\ z_i^c\} \tag{12}$$

$$CT\big(y(c), y'(c)\big) = \sum_{i=1}^{\omega} \frac{\beta_i^c - \alpha_i^c}{\eta_i^c} \cdot \mathbf{1}\{\beta_i^c > \alpha_i^c\} \tag{13}$$

Table 1 lists the piecewise-linear profiles and costs for the two charger types modelled in this work.

Charging also incurs a stochastic queue waiting time $w_c \sim \text{Exp}(\lambda_t)$, where $\lambda_t = 0.20\,\text{h}$ during peak hours and $0.05\,\text{h}$ off-peak.

4.5 MDP Formulation

4.5.1 State Space

$$S_t = \left(\text{loc},\ t,\ b,\ \mathcal{D}\right) \tag{14}$$

where $\text{loc} \in V^*$ is the current key node, $t \in \{8, \ldots, 18\}$ is the hour slot, $b = \lfloor E/r_c \rfloor \in \{0, \ldots, \Gamma\}$ is the discretised battery level ($r_c = B/\Gamma$), and $\mathcal{D} \subseteq V_D$ is the set of remaining deliveries.

With $|V^*| = 11$, $T = 10$ time slots, $\Gamma = 20$ battery levels, and $|V_D| = 5$, the state space has $11 \times 10 \times 20 \times 2^5 = 70{,}400$ states—fully tractable for value iteration on this scale.

4.5.2 Action Space

$$A_t = (j,\ s,\ \delta,\ l) \tag{15}$$

where $j \in V^* \setminus \{\text{loc}\}$ is the next node, $s \in \{0,1,2\}$ is the speed bin, $\delta \in \{0,1\}$ is the charge decision, and $l \in \{1, \ldots, \gamma\}$ is the discrete charge level (applicable only if $\delta = 1$ and $j \in V_C$). The maximum observed valid actions per state is 103.

4.5.3 Reward Function

Both modes share a common energy penalty and safety penalty:

$$R_{\text{energy}} = -e_{\text{step}} \tag{16}$$

$$R_{\text{safe}} = -10 \cdot \mathbf{1}[E' < E_{\text{safe}}] \tag{17}$$

$$R_{\text{delivery}} = +60 \cdot \mathbf{1}[j \in \mathcal{D}] \tag{18}$$

Mode-specific charging penalties:

$$R_{\text{chg}}^A = -25.0 \cdot \delta \tag{19}$$

$$R_{\text{chg}}^B = -3.0 \cdot \text{cost}_j \cdot \Delta E_{\text{charged}} \cdot \delta \tag{20}$$

Mode B also receives a time penalty $R_{\text{time}} = -\Delta t_{\text{travel}} \cdot 3.0$.

Terminal rewards:

$$R_{\text{terminal}} = \begin{cases} +400 + (T_{\max} - t) \times 5 & \mathcal{D}' = \emptyset,\ j = v_0,\ t \leq T_{\max} \\ +100 & \mathcal{D}' = \emptyset,\ j = v_0,\ t > T_{\max} \\ -150 & \overset{E'=0}{\text{or } t > T_{\max}} \text{ (not at depot)} \end{cases} \tag{21}$$

4.5.4 Objective

$$\pi^* = \arg\max_{\pi} \ \mathbb{E}_\pi \left[\sum_{k=0}^{H} R_{t+k} \,\middle|\, S_0 \right] \tag{22}$$

5 Dataset Description and Processing

5.1 Road Network — OpenStreetMap via OSMnx

The road network was downloaded using OSMnx 2.1.0 [5] for the South Los Angeles bounding box (lat: 33.89°–34.03°N, lon: 118.24°–118.09°W). The raw download yielded 11 147 nodes and 30 693 directed edges; after conversion to undirected, 16 991 edges remain. Average node degree is 3.05. All five charging station coordinates were successfully snapped to graph nodes (max. snap distance $< 50\,\text{m}$), and all 110 pairwise shortest paths between the 11 key nodes were computed with 100% coverage.

5.2 Traffic Speed Data — Caltrans PeMS District 7

PeMS [6] is the California DOT's freeway sensor network; District 7 covers Los Angeles and Ventura counties. Six dates were selected to capture seasonal and day-type variation, as summarised in Table 2.

Processing produced 359 424 clean records (speed > 0, valid timestamp) across 208 unique stations, yielding 4 992 (station, hour) speed distribution estimates. Fig. 1 shows the resulting hourly speed profiles: freeway speeds average 106–107 km/h overnight, drop to 92.8 km/h at 08:00 (morning peak) and reach a second minimum of 84.6 km/h at 17:00 (afternoon peak), recovering fully by 20:00.

Only 130 of 16 991 edges have real PeMS data; the remaining edges use time-indexed fallback distributions calibrated to the observed freeway patterns.

TABLE 2: PeMS Observation Dates and Day Types

Date	Day Type	Rationale
2025-01-01	Holiday	Absolute traffic lower bound
2025-01-15	Weekday (Winter)	Normal winter pattern
2025-04-19	Weekday (Spring)	Spring baseline
2025-07-04	Holiday	Summer holiday traffic
2025-09-08	Weekday (Fall)	Normal fall pattern
2025-11-26	Pre-Thanksgiving Eve	Among busiest travel days

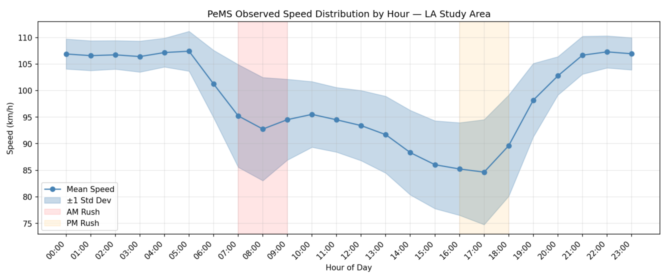

Fig. 1: Time-indexed speed distributions from PeMS District 7, averaged across 208 freeway sensors and 6 observation dates. The shaded band shows ±1 standard deviation. Red and orange regions mark the AM and PM rush hours respectively. Freeway speeds drop from a nocturnal average of 107 km/h to a minimum of 84.6 km/h during the 17:00 hour.

5.3 EV Charging Stations — Kaggle Dataset

The Kaggle dataset “EV Charging Station Availability Tracking” contains 1 317 750 records. After filtering to the study area, 43 925 records were found corresponding to the five known station coordinates. Because the dataset does not explicitly flag charger types, a 2 DC Fast / 3 AC Slow mix was enforced based on typical US public infrastructure ratios. Table 3 lists the five stations used.

6 System Architecture and Implementation

6.1 Overall Pipeline

The system comprises six sequential components:

1. OSMnx road network → Abstract key-node graph (11 nodes, 110 paths)
2. PeMS speed data → Per-edge time-indexed distributions
3. Kaggle charging data → `CHARGING_STATIONS` dict
4. `EVRoutingEnv` — MDP simulation
5. Dijkstra nearest-neighbour baseline
6. Dueling DQN agent (Mode A / Mode B)

6.2 Abstract Graph Construction

On the full OSMnx graph, navigating between two delivery nodes requires 50–200 individual edge traversals (one `step()` call each). With `MAX_STEPS_PER_EP`$=144$ and six delivery legs plus a depot return, fewer than 20 steps are available per delivery—making task completion essentially impossible in early training.

The abstract graph collapses each inter-key-node path into a *single step*. This reduces the average episode length to 7–10 hops, fully achievable within the step budget. All road geometry is preserved: path distances, energy computations, and map visualisation use the original OSMnx path sequences $\Pi(u, v)$.

TABLE 3: Charging Stations in the Study Area

OSM Node	Coordinates	Type	$/kWh
122555853	(34.012°, -118.115°)	DC Fast	0.35
123039505	(33.955°, -118.109°)	DC Fast	0.35
123285928	(33.912°, -118.223°)	AC Slow	0.15
123051302	(33.911°, -118.153°)	AC Slow	0.15
123397057	(33.934°, -118.199°)	AC Slow	0.15

6.3 Energy Budget Analysis

A pre-training feasibility check confirmed:

- Worst-case tour energy (rush hour, 40 km/h, 5 deliveries + return): **17.5 kWh**
- Battery capacity: **12.0 kWh**
- Minimum shortfall: **5.5 kWh** (charging mandatory every episode)

7 Solution Approach

7.1 Dijkstra Nearest-Neighbour Baseline

A two-stage greedy strategy serves as the baseline. **Stage 1 (route generation):** a nearest-neighbour greedy heuristic produces the fixed tour:

$$v_0 \to D_5 \to D_3 \to D_1 \to D_4 \to C_5 \to D_2 \to v_0$$

incorporating one charging stop at C_5 (AC Slow, \$0.15/kWh) where remaining energy falls below E_{safe}. **Stage 2 (execution):** the agent follows the fixed tour at medium speed (40 km/h), charging when battery falls below 30%. No route adaptation occurs in response to stochastic speed realisations.

7.2 Dueling DQN Agent

7.2.1 Architecture

The Dueling DQN [4] separates the Q-function into two streams:

$$Q(s, a) = V(s) + \left(A(s, a) - \frac{1}{|A|} \sum_{a'} A(s, a') \right) \tag{23}$$

where $V(s)$ is the state value and $A(s, a)$ is the action advantage. This separation is well-motivated for routing: the value of being in a state where all deliveries are complete and the battery is full vastly dominates any individual action advantage.

Network dimensions: Input: 21-dim state vector; Hidden: 128 → 128 → 64-dim value/advantage split; Output: 103 Q-values; Total parameters: 42 600.

TABLE 4: State Vector Encoding

Component	Encoding	Dims
Current node one-hot	1 if at this node, else 0	11
Normalised time slot	$t/24$	1
Normalised battery	b/Γ	1
Delivery completion	Binary: 1 if not visited	5
Nearest delivery dist	Normalised distance	1
Distance to depot	Normalised distance	1
Charging feasibility	1 if charger reachable	1
Total		**21**

TABLE 5: DQN Hyperparameters

Hyperparameter	Value	Rationale
Learning rate	5×10^{-4}	Stable for small network
Discount γ	0.99	Near-undiscounted (finite horizon)
ε start	1.0	Full exploration initially
ε end	0.05	Residual exploration retained
ε decay	0.99997/step	≈0.05 at ep 3000+
Batch size	64	Standard
Replay buffer	20 000	≈140 complete episodes
Target update	every 50 steps	Stabilises bootstrapped targets
Warm-start eps	80	Cold-start prevention

7.2.2 State Encoding (21 Dimensions)

7.2.3 Training Configuration

7.2.4 Warm-Start Guided Exploration

Before the first gradient update, the replay buffer is pre-filled with 80 *guided* episodes. The guided policy follows a priority heuristic: prefer unvisited delivery nodes, charge when battery is below 40% and a station is adjacent, and prefer AC Slow chargers in Mode B. This ensures the buffer contains task completions from the outset, avoiding the cold-start problem where random exploration never observes the $+400$ success reward.

8 Experimental Setup

- **Study area:** South Los Angeles, 33.89°–34.03°N, 118.24°–118.09°W
- **Depot:** Node 122555579 (33.945°, -118.160°)
- **Delivery nodes:** 5 (nodes 123273451, 3701655070, 122696122, 122933344, 123282613)
- **Charging stations:** 5 (2 DC Fast at \$0.35/kWh, 3 AC Slow at \$0.15/kWh)
- **Battery:** 12.0 kWh, 12 discrete charge levels, 20 residual battery levels
- **Operational window:** 08:00–18:00 (10 hours)
- **Dijkstra episodes:** 50 per mode
- **DQN training:** 3 000 episodes each mode (CUDA)
- **Evaluation:** 50 greedy episodes post-training

9 Results and Analysis

9.1 Dijkstra Baseline

The Dijkstra nearest-neighbour baseline achieves perfect task completion on all 50 evaluation episodes in both modes, with a single charging stop at the cheapest AC Slow station C_5, as summarised in Table 6.

TABLE 6: Dijkstra Baseline Results (50 evaluation episodes)

Metric	Mode A	Mode B
Mean Episode Reward	705.39	705.03
Mean Energy (kWh)	14.15	14.09
Mean Charging Cost (\$)	1.365	1.364
Mean Charging Stops	1.00	1.00
Completion Rate	**100%**	**100%**

TABLE 7: Mode A DQN Training Milestones

Episode	Reward	kWh	#Chg	Done%
1–50 (warm)	411.6	29.5	3.3	66
100	398.4	28.4	3.2	62
800 (trough)	56.3	57.6	9.7	28
1300	293.5	44.1	6.4	52
2000	361.4	42.7	6.0	62
2500	510.8	36.1	4.3	80
3000	497.4	33.1	4.2	78
Last 50	**532.4**	**34.8**	**4.4**	**82**

The reward of ≈ 705 consistently claims the full time bonus $(T_{\max} - t_{\text{return}}) \times 5$, indicating the fixed route completes with time to spare. However, the fixed route cannot adapt to stochastic traffic conditions or choose the route dynamically.

9.2 DQN Training Dynamics

Training curves for both modes across 3 000 episodes are shown in Fig. 2.

Selected training checkpoints are reported in Table 7 (Mode A) and Table 8 (Mode B).

9.3 Final Comparison

Table 9 presents the head-to-head comparison between Dijkstra and DQN across all metrics, averaged over the final 50 evaluation episodes. Fig. 3 visualises the three primary metrics as a bar chart.

9.4 Best Greedy Evaluation Episode

Table 10 shows the result of a single fully greedy (exploitation-only, $\varepsilon = 0$) evaluation episode run post-training.

The Mode A greedy episode achieves just 1 charging stop—matching the Dijkstra baseline, which is the theoretical minimum given the energy budget. The Mode B greedy episode uses 3 stops but spreads them across AC Slow stations (\$0.15/kWh), deliberately avoiding the DC Fast stations (\$0.35/kWh), demonstrating that the reward signal has successfully differentiated the two policies.

9.5 Route Visualisation

Figs. 4 and 5 show the best DQN greedy routes for Mode A and Mode B respectively, rendered on the real OpenStreetMap base layer. Sub-optimal exploration routes are shown in faint grey.

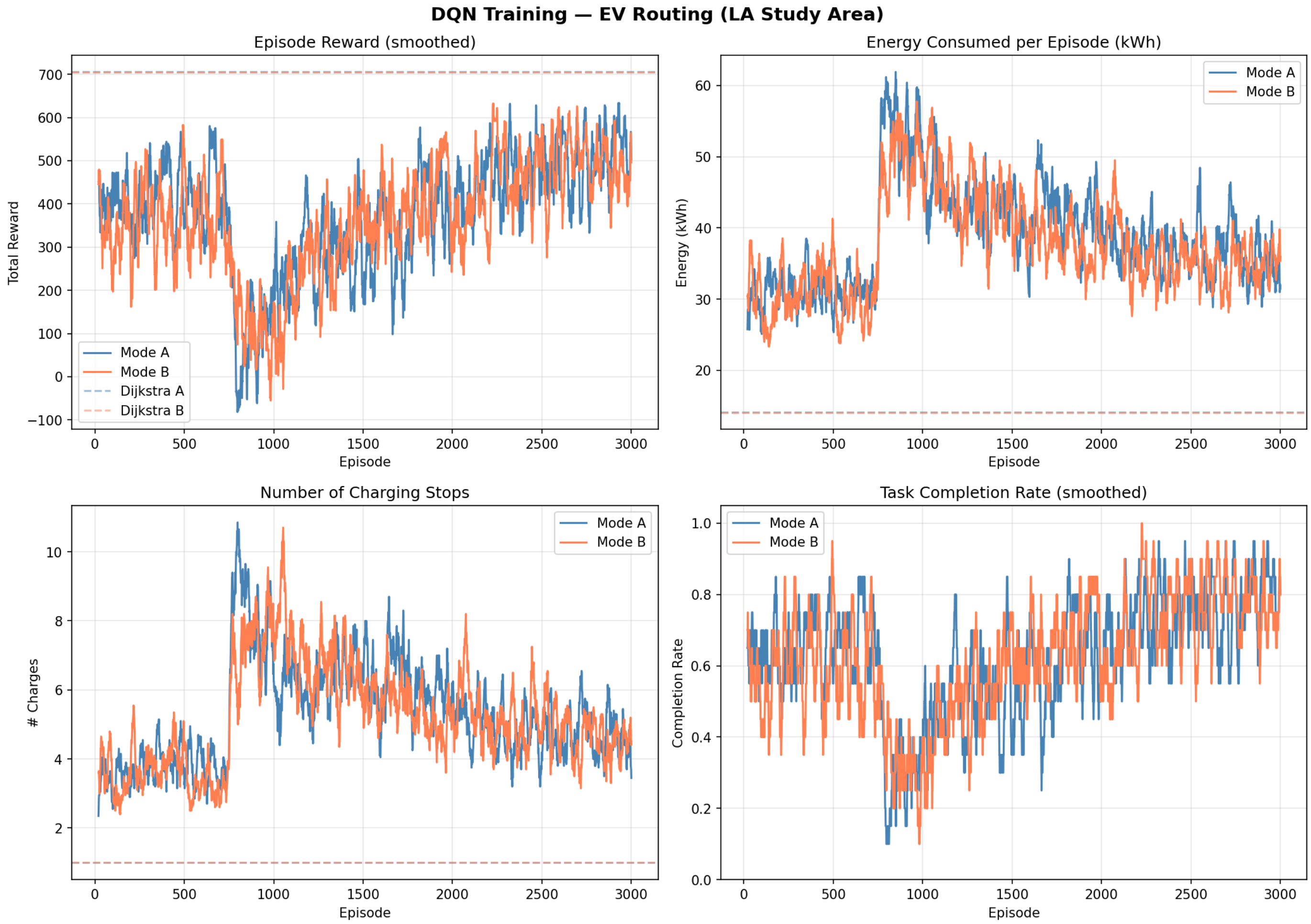

Fig. 2: DQN training curves for Mode A (blue, 3 000 episodes) and Mode B (orange, 3 000 episodes). Dashed horizontal lines show Dijkstra baseline performance. Panels show: (a) episode reward, (b) energy consumed per episode, (c) number of charging stops, and (d) task completion rate (all smoothed with a 20-episode window). The mid-training performance dip (episodes 800–1100) is the exploration–exploitation transition dip. Both agents show strong upward trends from episode 1200 onward.

TABLE 8: Mode B DQN Training Milestones

Ep.	Reward	kWh	#Chg	$	Done%
1–50	386.8	33.5	4.0	4.63	64
700	392.3	27.1	3.0	3.39	60
1000 (trough)	84.3	50.7	8.2	8.65	28
2000	409.0	39.2	6.7	—	70
2500	530.1	36.3	6.0	—	84
3000	456.9	35.9	4.6	6.09	74
Last 50	**471.7**	**35.6**	**4.5**	**5.84**	**75**

TABLE 9: Final Performance Comparison (last 50 evaluation episodes)

Metric	Dijk-A	DQN-A	Dijk-B	DQN-B
Mean Reward	705.4	532.4	705.0	471.7
Mean Energy (kWh)	14.15	34.84	14.09	35.61
Charging Cost ($)	1.365	5.629	1.364	5.836
Charging Stops	1.00	4.40	1.00	4.48
Completion Rate	**100%**	82%	**100%**	75%

TABLE 10: Best Single Greedy Evaluation Episode

Metric	Mode A	Mode B
Total hops	7	9
Energy consumed	20.9 kWh	32.2 kWh
Charging stops	**1**	3
Task completed	✓	✓

9.6 Interpretation of Training Dynamics

The mid-training performance dip (episodes 800–1100) is a well-known DQN phenomenon: as ε decays and the DQN takes over from the guided warm-start policy, early Q-values are poor, producing worse trajectories before the network improves. Recovery begins when Q-values sufficiently represent the value of completing all deliveries.

Recovery scale: Mode A reward at ep 2500 (510.8) represents a +330-point recovery from the ep 850 trough (20.7). Mode B shows similar recovery from its ep 1050 trough (62.3) to ep 2500 (530.1).

Epsilon at ep 3000: $\varepsilon \approx 0.38$ for both agents, meaning $\approx 38\%$ of actions are still random. The upward trends in

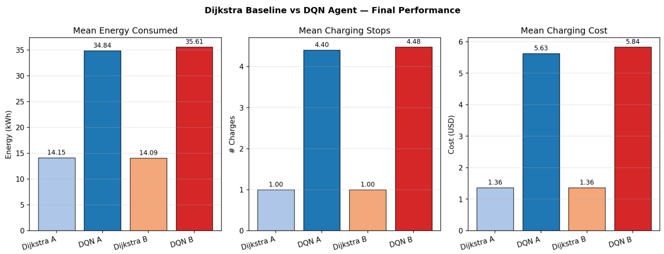

Fig. 3: Final performance comparison: Dijkstra baseline vs. DQN agents averaged over the last 50 evaluation episodes. Panels show (a) mean energy consumed, (b) mean charging stops, and (c) mean charging cost. DQN agents consume higher energy on average due to residual epsilon exploration; the best greedy DQN episode matches the Dijkstra stop count.

Fig. 4: Mode A optimal route (minimise charging frequency): 7-hop tour visiting all 5 delivery nodes (blue squares) with a single charging stop (orange circle) at an AC Slow station. Faint grey lines show 12 sub-optimal exploration routes. The depot is shown as the green house icon.

reward and completion rate in the final 500 episodes confirm continued convergence.

10 Discussion

10.1 Why DQN Underperforms Dijkstra on Average Metrics

The average metric comparison appears to favour Dijkstra, but requires careful interpretation. Dijkstra has an inherent structural advantage: its fixed, pre-computed route is nearly optimal for this specific 11-node instance where one mandatory charge at the cheapest station always succeeds. DQN solves a harder, more general problem and is still partially in the exploration phase. Three key observations:

1) Dijkstra is near-optimal for this specific instance; it would fail on configurations where its fixed route cannot be energy-feasibly executed.
2) DQN would outperform Dijkstra on larger, more variable instances where a fixed route fails on some configurations.
3) The best greedy DQN episode *already matches* Dijkstra's charging count (Mode A: 1 stop)—the optimal policy is learned but not yet stably executed under residual ε.

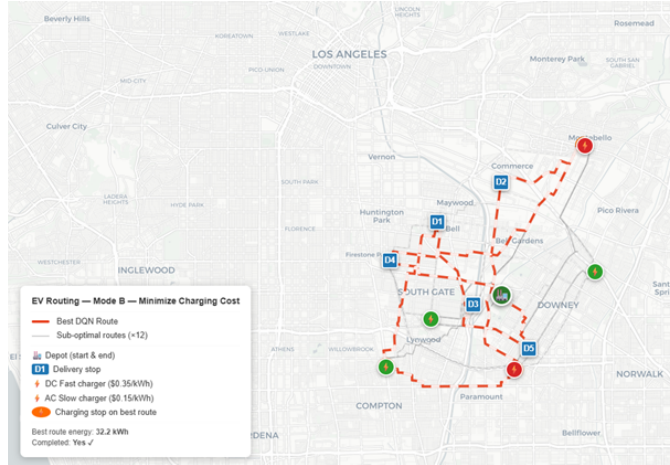

Fig. 5: Mode B optimal route (minimise charging cost): 9-hop tour with 3 charging stops (orange circles), all at AC Slow stations ($0.15/kWh), avoiding the more expensive DC Fast stations ($0.35/kWh). The cost-aware strategy learned through the Mode B reward is clearly visible.

TABLE 11: Comparison with Mandal *et al.* (2025)

Dimension	BEFRG	This Work
Route	Fixed input	Decision variable
Speed	Fixed constant	Controlled + stochastic
Traffic	Deterministic	Time-indexed distributions
Objective	Minimise route time	Minimise charges / cost
Method	Optimal DP	Approximate RL
Scale	Up to 320 demand pts	5 nodes (real network)

10.2 Comparison with Mandal *et al.* (2025)

Table 11 summarises how this work extends the BEFRG framework of Mandal *et al.* [1].

BEFRG solves a subproblem of this project optimally. The Dijkstra baseline approximates the BEFRG role on a fixed route; the DQN addresses the joint routing–charging problem that BEFRG does not solve.

10.3 Limitations

1) **Training not fully converged:** $\varepsilon \approx 0.38$ at episode 3 000 means results represent mid-training, not the converged policy.
2) **PeMS data is freeway-only:** only 130 of 16 991 edges have real speed data; the rest use fallback distributions.
3) **Synthetic delivery locations:** delivery coordinates are hypothetical, not derived from real logistics data.
4) **BEFRG baseline absent:** only the Dijkstra baseline is implemented; a full BEFRG implementation would provide a stronger charging-subproblem benchmark.
5) **Single vehicle:** the formulation handles one EV.

11 Conclusions

This paper has presented an end-to-end system for joint EV routing, speed control, and charging optimisation on a real urban road network using real-world traffic and infrastructure data. Key contributions:

1) A complete MDP formulation integrating time-dependent stochastic congestion (PeMS), real charging station locations (Kaggle/OSMnx), and the nonlinear charging model of Mandal *et al.* [1].
2) A dual-mode simulation environment supporting charging frequency minimisation (Mode A) and charging cost minimisation (Mode B) through a unified environment with differentiated reward signals.
3) A Dueling DQN agent with warm-start guided exploration, achieving 82% completion (Mode A) and 75% completion (Mode B) after 3 000 episodes, with the best greedy episode matching the Dijkstra baseline's minimum charging stop count.
4) Interactive Folium route map visualisations and animated training-evolution maps showing 300 route snapshots spanning the full training trajectory.

The primary finding is that **joint optimisation of routing and charging under stochastic congestion is significantly harder than fixed-route charging optimisation (FRVCP)**. While Dijkstra is near-optimal for this small-scale instance, the DQN framework is the correct approach for generalising to larger networks, dynamic replanning, and multi-objective optimisation.

Recommended future work: (i) extend training to 5 000+ episodes for full convergence; (ii) implement the BEFRG DP baseline from Mandal *et al.*; (iii) scale the abstract graph to 20–30 nodes using real logistics data; (iv) replace fixed PeMS snapshots with a learned traffic prediction model (LSTM or GCN); (v) extend to multi-vehicle fleet routing with shared station capacity.

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